Higher-Order Linear Differential Equations:-

are the methods of solution that are readily extended to an nth-order linear differential equation. With constant coefficients and a constant term, such an equation can be written ~~as~~ generally as:

$$y^{n}(t) + a_1 y^{n-1}(t) + \cdots + a_{n-1} y'(t) + a_n y = b.$$

Solution of the HOLDE.

In higher order because we have nth-order linear differential equation with constant coefficient Eq. constant term.

$$\therefore \quad y_p = \frac{b}{a_n} \quad \text{when } \boxed{a_n \neq 0.}$$

and $$y_p = \frac{b}{a_{n-1}} t \quad \text{when } \boxed{a_n = 0 \text{ but } a_{n-1} \neq 0.}$$

While $y_p = Kt^2$ when $\boxed{a_n = a_{n-1} = 0.}$

on the other hand

$$y_c = A_1e^{r_1t} + A_2e^{r_2t} + \dots + A_ne^{r_nt}.$$

when roots are $\boxed{\text{real \& distinct}}$

and

$$y_c = A_1te^{r_1t} + A_2t^2e^{r_1t} + \dots + A_nt^ne^{r_1t}.$$

when roots are $\boxed{\text{Repeated}}$

while

$$y_c = e^{ht}(A_1\cos vt + A\sin vt).$$

when roots are $\boxed{\text{complex.}}$

The General Solution HOLDE is

~~$y_t = y_p + y_c$.~~ $y_t = y_p + y_c$.

y_p = Particular integral.

y_c = Complementary integral.

while $y_p = Kt^2$ when $\boxed{a_n = a_{n-1} = 0}$

Convergence and the Routh theorem:-

The solution of a higher-degree characteristic equation is not always an easy task. For this the routh theorem become helpful for the convergence or divergence of a time path without having to solve for the characteristic roots. Which can provide a qualitative analysis of a differential equation.

$$|a_1| ; \begin{vmatrix} a_1 & a_3 \\ a_0 & a_2 \end{vmatrix} ; \begin{vmatrix} a_1 & a_3 & a_5 \\ a_0 & a_2 & a_4 \\ 0 & a_1 & a_3 \end{vmatrix} ; \begin{vmatrix} a_1 & a_3 & a_5 & a_7 \\ a_0 & a_2 & a_4 & a_6 \\ 0 & a_1 & a_3 & a_5 \\ 0 & a_0 & a_2 & a_4 \end{vmatrix} ; \ldots$$

we always take $a_0 = 1$.
If we have third degree polynomial equation $(n=3)$, we should set $a_4 = a_5 = a_6 = a_7 \dots = 0$.

Higher Order linear difference equations and their Solutions:-

The order of a difference equation indicates the highest-order difference present in the equation; but it also indicates the maximum number of periods of time lag involved. An nth-order linear difference equation may thus be written in general as:

$$y_{t+n} + a\, y_{t+n-1} + \dots + a_{n-1} y_{t+1} + a_n y_t = c.$$

$\Delta y_t = 0.$

$\Delta y_t = y_{t+1} - y_t$

$$y_{t+1} - y_t = 0.$$

$$\boxed{y_p = k = \frac{c}{1+a}}$$

$$\boxed{y_c = \left(y_0 - \frac{c}{1+a}\right)(-a)^t}$$

$$y_t = y_c + y_p$$

OR.

$$\boxed{y_t = \left(y_0 - \frac{c}{1+a}\right)(-a)^t + \frac{c}{1+a}}$$

Convergence and the Schur Theorem:-

The solution of higher-order difference equation is not always an easy task. For this the schur theorem become helpful for the convergence or divergence of a time path with out having to solve for the characteristic roots. Which can provide a qualitative analysis of a difference equation.

$$\Delta_1 = \begin{vmatrix} a_0 & a_2 \\ a_2 & a_0 \end{vmatrix} ; \quad \Delta_2 = \begin{vmatrix} a_0 & 0 & a_2 & a_1 \\ a_1 & a_0 & 0 & a_2 \\ a_2 & 0 & a_0 & a_1 \\ a_1 & a_2 & 0 & a_0 \end{vmatrix} .$$